Element D

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The solutions that we came up with have been thoroughly evaluated by having the ideas of materials being used, cost of materials, as well as specific attributes that have been deemed as necessary by Dr.Behnam (main stakeholder) as well as our other stakeholders in the students at Drexel University. The solutions shown below are based on many qualities of designs that our group members have agreed upon to be implemented to the surgical basin. Along with our designs below, there is also a functions chart which shows the functions of our basins, as well as six different solutions that can be used in order to treat the function. There is also a weighted decision matrix with qualities of designs that are the most important having the largest weight in choosing our final design.

Functions Chart

Function	Solution 1	Solution 2	Solution 3	Solution 4	Solution 5	Solution 6
Form Fit	Contoured edges	Cut-out with foam	Basin made of malleable material	Cushion on top	Multiple size options	Net/sling
Resist heat (Autoclaving)	Stainless Steel (304)	Polypropylene	Polyetheretherk etone	Autoclavable silicone	Polysulfone	Fluoropolymer (Teflon)
Non-slip	Rubber coated bottom	Rubber feet	Suction cups	Weighted base	Textured bottom	Attached to table
Anti-spill	Wide base	Weighted base	Clamping mechanism	Suction cups	Curved sides	High Rised rim
Hold Liquid	Walled basin	Round edged bowl	Deep reservoir	Funnel-shaped base	Sloped bottom to channel fluid	Higher side walls
Adaptable	Multiple different sizes	Interchangeable parts	Expands and contracts	Malleable basin walls	Wide funnel-type rim	Rigid and unadjustable
Comfort for limb support	Medical grade foam padding	Silicone gel pad	Memory-foam contour	Inflatable soft cuff	Soft rubber arm rest	Neoprene sleeve insert
Lightweight/ portable	Thin-walled polypropylene	Aluminum alloy	Composite polymer	Minimalist structure	Foldable design	Nested stacking design
Durability/ structural strength	Reinforced rim	Cross-ribbed underside	Thick-gauge material	Impact-resisten t polymer	Double-layer walls	Metal-polymer hybrid
Secure patient limb positioning	Slight arm/leg groove	Adjustable strap	Rubbersized grip area	Elevated center ridge	Soft clamp mechanism	Form stabilizer block

Weighted Decision Matrix

		Design #1		Design #2		Design #3		Design #4		Design #5		Design #6		Design #7	
Qualities of Design	Weight	Rating	Value	Rating	Value	Rating	Value	Rating	Value	Rating	Value	Rating	Value	Rating	Value
Ergonomic Comfort/Fit	20%	4	0.80	4	0.80	4	0.80	3	0.60	2	0.40	3	0.60	2	0.40
Stable Positioning	20%	3	0.60	4	0.80	3	0.60	4	0.80	1	0.20	1	0.20	3	0.60
Sterility/Sanitation	20%	3	0.60	3	0.60	3	0.60	3	0.60	3	0.60	3	0.60	2	0.40
Safety	10%	3	0.30	3	0.30	2	0.20	2	0.20	3	0.30	3	0.30	2	0.20
Cost Effective	10%	2	0.20	3	0.30	2	0.20	2	0.20	2	0.20	1	0.10	1	0.10
Spill Resistant	10%	2	0.20	3	0.30	2	0.20	3	0.30	4	0.40	4	0.40	3	0.30
Lightweight/Portable	10%	3	0.30	3	0.30	3	0.30	1	0.10	1	0.10	1	0.10	2	0.20
Total	**100%**		**2.70**		**3.10**		**2.60**		**2.70**		**2.10**		**2.20**		**2.00**

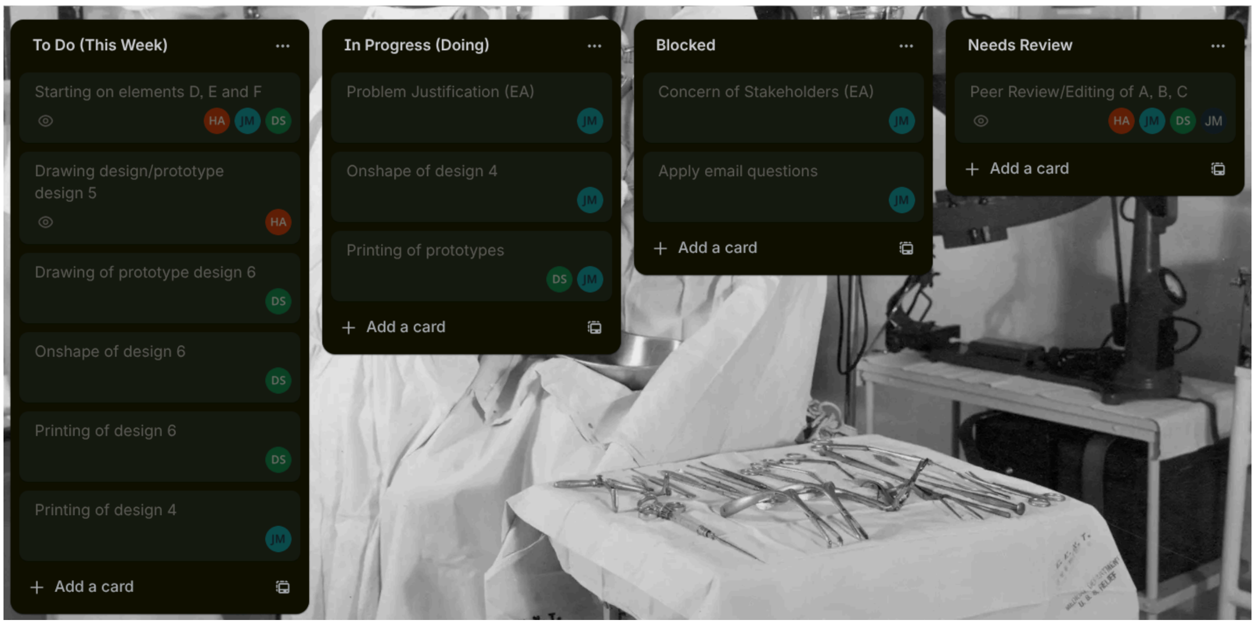
The Decision:

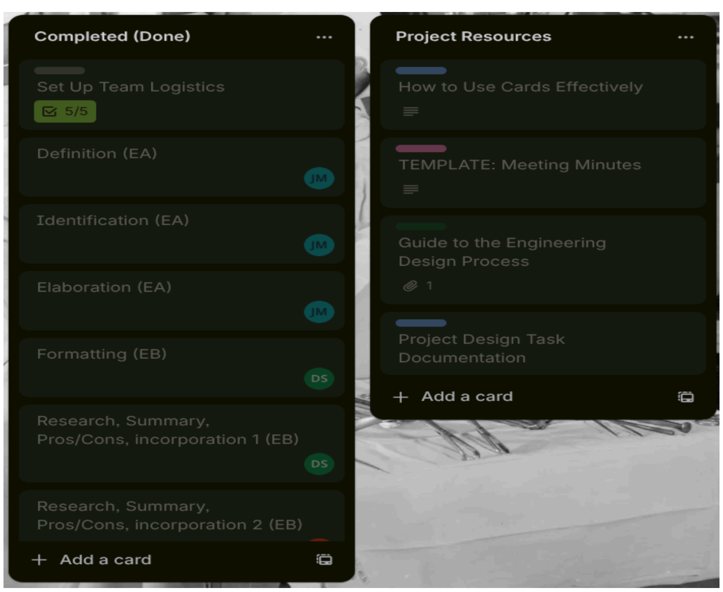
After the meeting with Dr.Behnam and the medical students at Drexel, we all came to a consensus of a prototype that is scaled down ¼ of the dimensions and 3d printed. The prototype is a basin that is 3d printed, with defects/u-shape on all four sides, as well as some foam. The main objective of this project is to design, test, and build an emesis basin that is comfortable and is a simple easy to follow design. The u-shape on all four sides of the basin allows for arms to be placed, giving the patient more comfort than the traditional basin that is used.

After careful consideration, our group has locked in on Design #2 as the solution to the basin. This is due to the fact that the basin allows for the most amount of versatility, design simplicity, low cost, as well as allowing for ergonomic comfort, stable positioning, and sterility which were all top design requirements for us. Design #2 was the clear stand out due to its ability of being lightweight which the majority of other designs were not. The simplicity of the design also allows for quicker construction, which allows for more testing time in order to put out the best product possible. Finally, Design #2 was the only design that scored a 3 or a 4 on every single design requirement according to the weighted decision matrix.

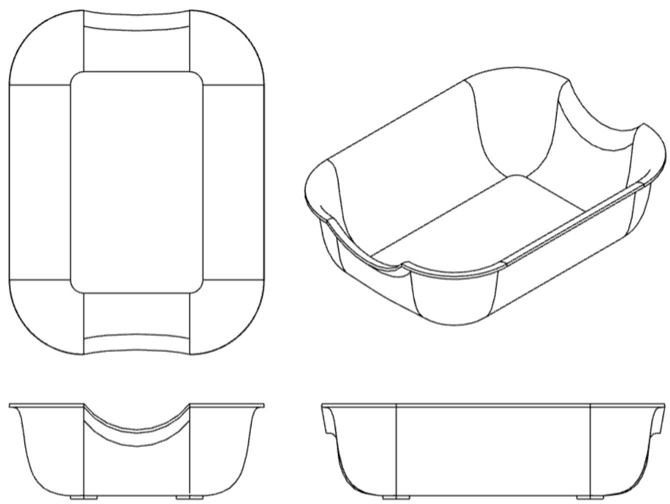
Before all of the designs are presented, below is attached a screenshot of the trello that has been used throughout the course of this project so far. The trello allows for us to set due dates, as well as to assign each other tasks throughout, and to keep each group member in check. This allows for more efficient work to be done, as well as allowing adequate time for the project to be completed, and most importantly, to lead to the best possible product to be put out.

Trello



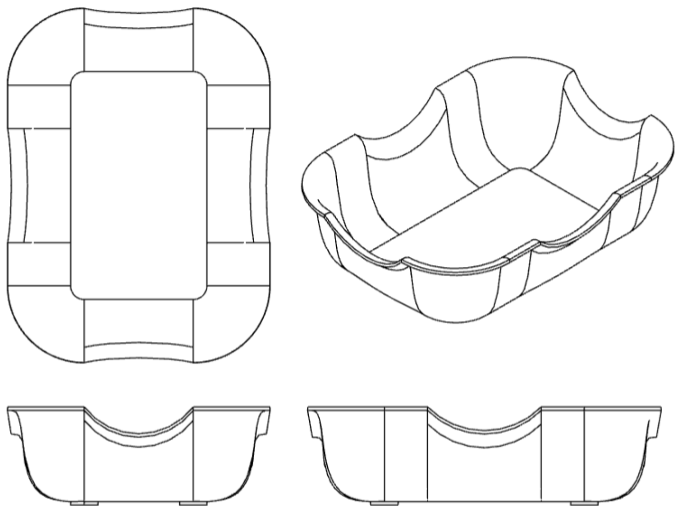


Design #1



This design was our first idea, and it was created based on Dr. Behnam's original problem. It includes two equally sized defects on two sides of the basin. It would be manufactured using an injection mold with polypropylene. It features four circular feet on the bottom to avoid slipping. High friction attachments can be added to the feet to avoid slipping.

Design #2

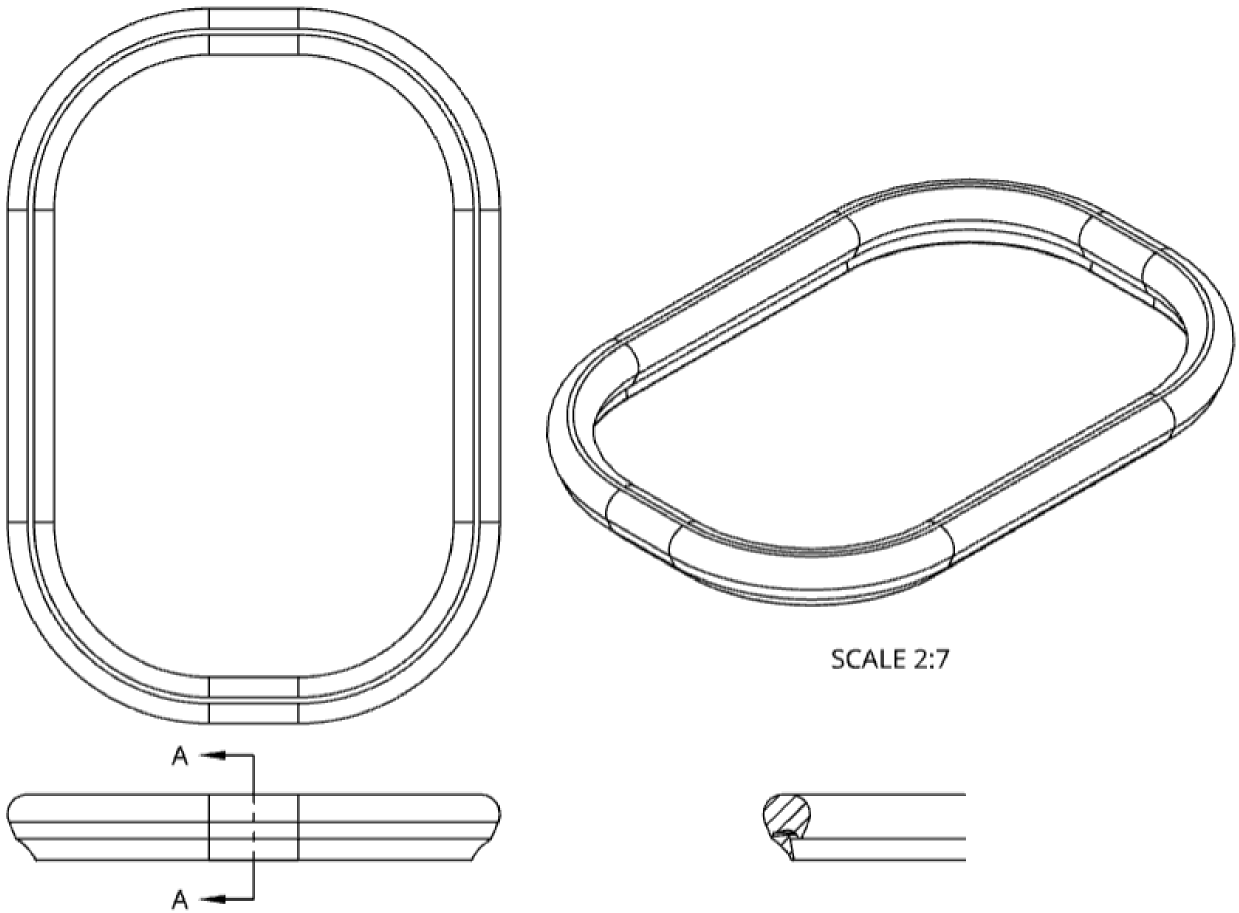


This design is similar to concept #1 with similar aspects such as the feet and the size, except it includes four defects on all four sides of the basin. This would allow surgeons or other medical professionals to place a bent limb on top of the basin, making the design more practical and applicable to a medical setting.

Design #3

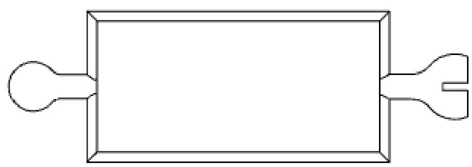
This design is also similar to concept #1 except it includes a foam cushion on the defect, securing the arm in place and increasing the comfort of the patient. This design has the ideal level of comfort for the patient, however, there are concerns over sterility and materials viability and cost.

Design #4

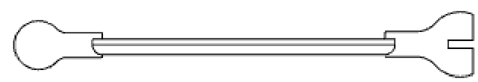


This design is completely different from the others. It is a silicone lip that attaches to the outside of an ordinary basin, giving the patients a secure cushion, while eliminating the need for a full basin redesign. There are concerns over the ability to create a universal lip that will attach to most basins as well as the design's ability to stay attached securely during surgery.

Design #5



SCALE 1.2:1



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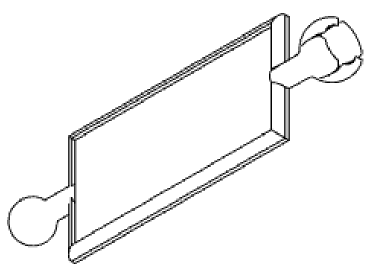
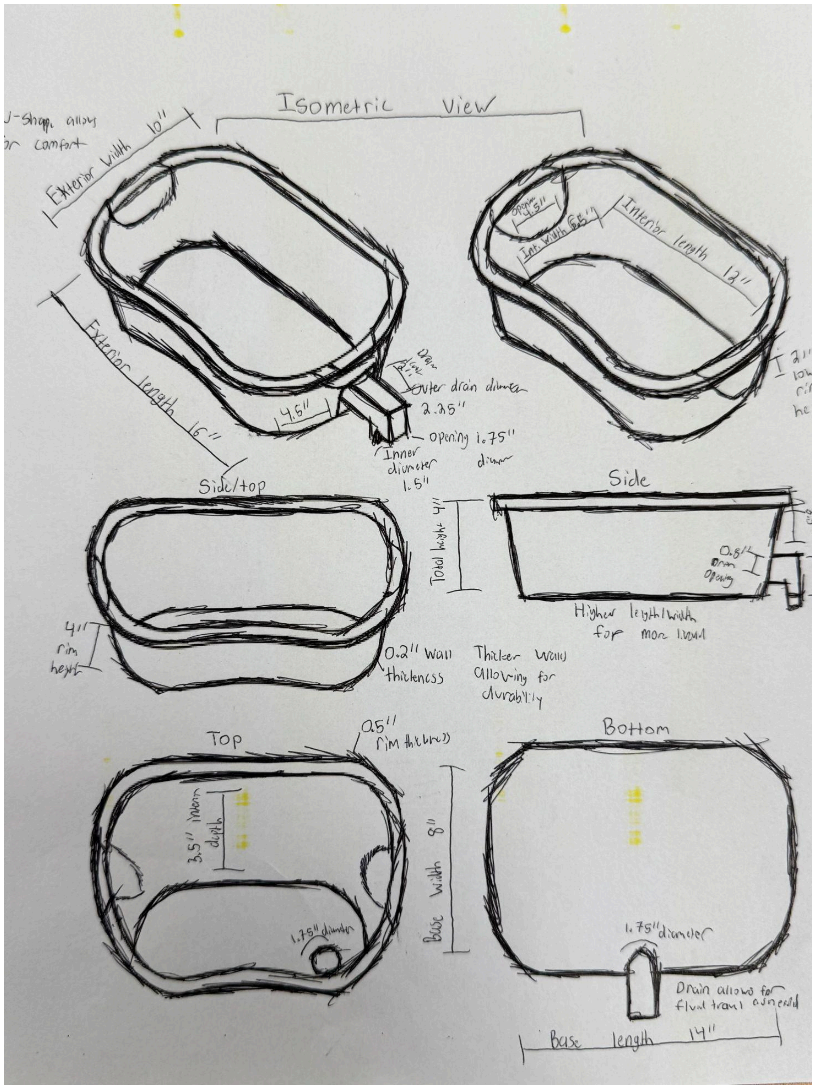


SCALE 1.2:1

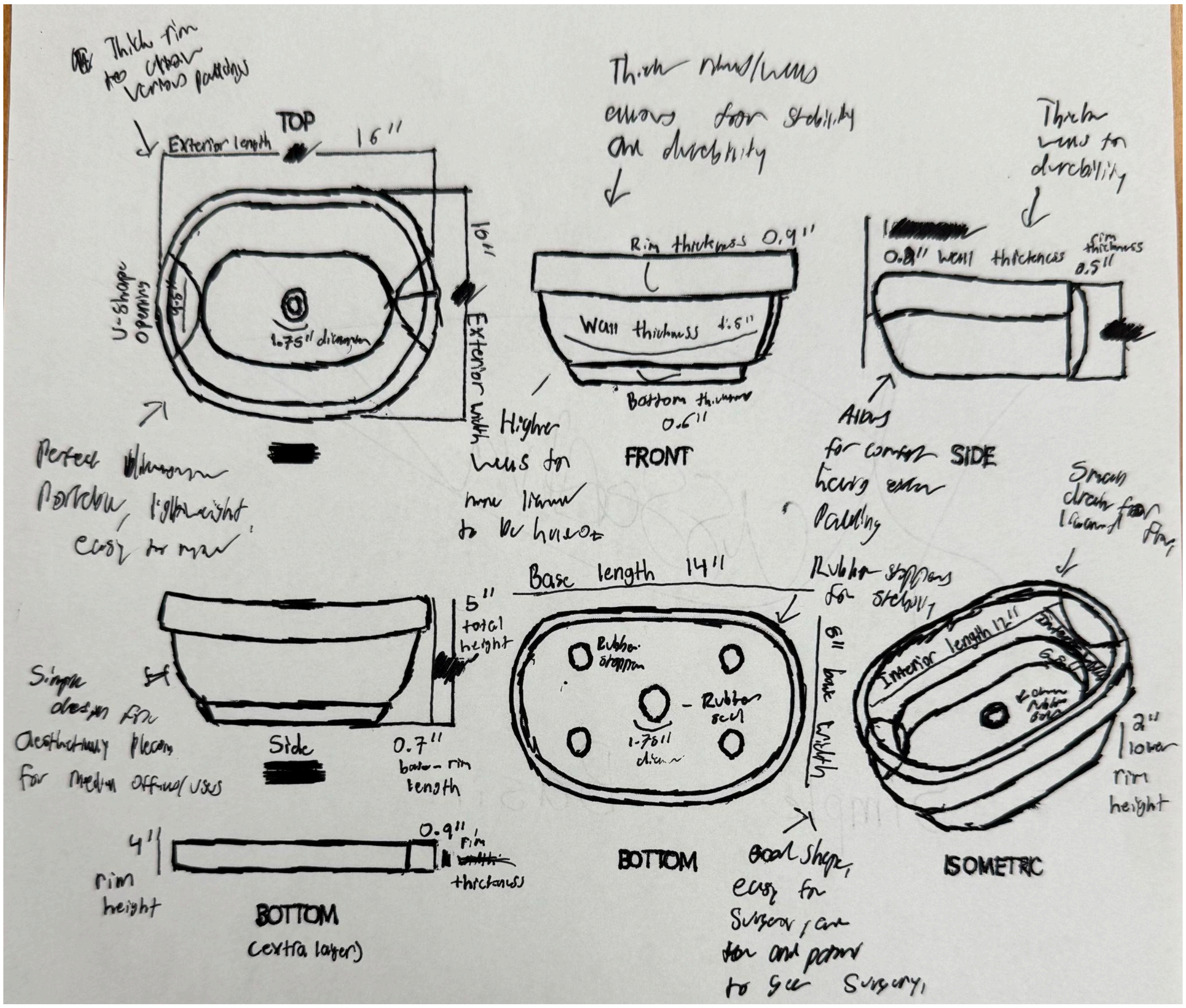
The Vertebrae System has 4-6 articulated segments connected by ball-and-socket joints that snap together and wrap around existing basin rims. Each segment has a rigid polypropylene base with a soft foam cushion on top to prevent patient bruising. The segments flex independently to conform to any basin shape (kidney, round, or rectangular), eliminating the need for multiple basin types. Suction cups or magnetic closures secure the attachment in place, providing stable limb positioning during irrigation procedures.

Design #6



A cradle-shaped support basin designed to securely hold a limb while giving the user a comfortable way to place their arm/leg into it. It is shaped to support the limb comfortably using angled walls. It prevents rolling or slipping because of the tall rim. It also allows for easy entry/exit with the -u-shape, it also provides a stable bottom so the limb stays in one position. Potentially allows for padding to be added. Valve on the right side of the basin with a small diameter to allow efficient fluid drainage, stability, and ease for the surgeon to transfer liquids.

Design #7



A kidney-profile basin with a u-shape to allow for an arm or leg to rest comfortably, plus a removable foam cuff that wraps around the limb to stabilize it, as well as a tilted channel floor that directs fluid to a single low-point drain at the center of the basin. There are rubber stoppers not only on the drain, as well as around the bottom to allow for less movement during surgery which decreases the risk of spillage. The basin also includes a removable soft liner for comfort and an integrated universal clamp tab for mounting.